

Probability Readiness / Waiver Test

for a Berkeley-style Stat 134 course

Purpose. This test is meant to identify students who are already fully comfortable with the main material of a rigorous calculus-based undergraduate probability course. If you can solve about **16–18 of the 20 problems correctly, cleanly, and without outside help**, you are probably already prepared to place out of the course or move directly to more advanced material.

Instructions. Work carefully and justify any nontrivial claim. You may use standard facts from calculus and basic linear algebra. Answers without reasoning may be insufficient, even when numerically correct.

1. A fair coin is tossed 10 times. Compute

$$\mathbb{P}(\text{exactly 6 heads} \mid \text{at least 5 heads}).$$

2. A disease has prevalence 0.01. A test is positive with probability 0.97 for a diseased person and 0.04 for a healthy person. Compute the posterior probability of disease given a positive test.
3. Give an explicit example of three events that are pairwise independent but not mutually independent.
4. An urn contains 7 red balls and 5 blue balls. Three balls are drawn without replacement. Let X be the number of red balls drawn. Find $\mathbb{E}[X]$ without first deriving the full distribution of X .
5. Let $X \sim \text{Geom}(p)$ on $\{1, 2, 3, \dots\}$. Prove the memoryless identity

$$\mathbb{P}(X > m + n \mid X > m) = \mathbb{P}(X > n).$$

6. Let X have density $f_X(x) = 2x$ on $0 < x < 1$. Find the density of $Y = X^2$.
7. Let (X, Y) have joint density

$$f(x, y) = 2, \quad 0 < y < x < 1,$$

and 0 otherwise. Find the marginal density of X and compute $\mathbb{E}[Y \mid X = x]$.

8. Suppose $B \sim \text{Ber}(1/3)$ and, conditional on B , the random variable X satisfies

$$X \mid (B = 0) \sim \mathcal{N}(0, 1), \quad X \mid (B = 1) \sim \mathcal{N}(2, 1).$$

Compute $\mathbb{E}[X]$ and $\text{Var}(X)$ using conditioning.

9. Let X be a nonnegative integer-valued random variable. Prove the tail-sum identity

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} \mathbb{P}(X \geq k).$$

10. Let $X_1, \dots, X_n$ be i.i.d. with mean μ and variance $\sigma^2 < \infty$. Compute $\text{Var}(\bar{X}_n)$, where

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

11. State the central limit theorem for i.i.d. random variables with finite variance.

12. Let $S_n \sim \text{Bin}(n, 0.4)$. Using a normal approximation with continuity correction, approximate

$$\mathbb{P}(S_{200} \leq 70).$$

You may leave the answer in terms of Φ .

13. A system has 4000 independent components, each failing during a day with probability 0.0005. Use a Poisson approximation to estimate the probability of exactly one failure during the day.
14. Let $\{N(t) : t \geq 0\}$ be a Poisson process with rate λ . What is the distribution of the waiting time to the third arrival?
15. Conditional on the event $\{N(5) = 3\}$ for a rate- λ Poisson process, describe the joint distribution of the three arrival times in $(0, 5)$.
16. Consider the Markov chain on $\{0, 1\}$ with transition matrix

$$P = \begin{pmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{pmatrix}.$$

Find its stationary distribution.

17. For simple symmetric random walk on $\{0, 1, 2, 3, 4, 5\}$ absorbed at 0 and 5, compute the probability of hitting 5 before 0 when starting from 2.
18. In the same chain as Problem 17, compute the expected time to absorption when starting from 2.
19. A Galton–Watson branching process starts from one ancestor. Each individual has either 0 children or 2 children, with probabilities 0.4 and 0.6, respectively. Compute the extinction probability.
20. Let $S_n = X_1 + \cdots + X_n$ where the X_i are i.i.d. with

$$\mathbb{P}(X_i = 1) = \mathbb{P}(X_i = -1) = \frac{1}{2}.$$

Show that $M_n = S_n^2 - n$ is a martingale with respect to the natural filtration.

Instructor note. A strong performance should be judged by correctness, clarity, and fluency. Students who solve only the computational items but struggle on the structural items (conditioning, CLT statement, Markov chain reasoning, martingale verification) are usually still missing important pieces of the course.